

# Shiv Nadar University Chennai

|                     |                                               |             |
|---------------------|-----------------------------------------------|-------------|
| Degree & Branch     | B.Tech Artificial Intelligence & Data Science | Semester V  |
| Subject Code & Name | CS3807 – Deep Learning Laboratory             | AY: 2026–27 |

## Experiment 1

### Implementation of a Single Layer Perceptron for Binary Classification

#### 1. Objective

The objective of this experiment is to understand the concept of an artificial neuron and implement a Single Layer Perceptron from scratch for binary classification. Students will learn the perceptron learning algorithm, understand the importance of activation functions, visualize the learning process, and evaluate the classifier using a real-world dataset.

#### 2. Tasks to be Performed

1. Download and understand the dataset.
2. Perform Exploratory Data Analysis (EDA).
3. Preprocess the dataset.
4. Implement a Single Layer Perceptron from scratch.
5. Train the model using the perceptron learning rule.
6. Evaluate the classifier using standard performance metrics.
7. Analyze the effect of different learning rates.
8. Compare the implementation with Scikit-learn's Perceptron (optional).
9. Perform the perceptron learning algorithm for OR, NOT, and AND gates using Python.

#### 3. Background Theory

An Artificial Neuron is the fundamental building block of a neural network. It receives one or more input features, multiplies them by corresponding weights, adds a bias term, and applies an activation function to determine the output.

The Single Layer Perceptron is the earliest supervised learning algorithm capable of solving linearly separable binary classification problems.

## Mathematical Formulation

Weighted Sum

$$z = \mathbf{w}^T \mathbf{x} + b$$

where

- $\mathbf{x}$  : Input vector
- $\mathbf{w}$  : Weight vector
- $b$  : Bias
- $z$  : Linear combination

Prediction

$$\hat{y} = f(z)$$

Weight Update Rule

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$$

Bias Update

$$b_{new} = b_{old} + \eta(y - \hat{y})$$

where  $\eta$  denotes the learning rate.

## 4. Activation Functions

An activation function determines the output of a neuron based on the weighted sum of its inputs. It introduces non-linearity, enabling neural networks to learn complex patterns. Although the perceptron uses only the Step Activation Function, modern deep learning models employ differentiable activation functions for efficient optimization through backpropagation.

### Step Activation Function

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

The Step Activation Function produces a binary output and is therefore suitable only for linearly separable binary classification problems.

| Activation | Output Range        | Typical Applications                    |
|------------|---------------------|-----------------------------------------|
| Step       | $\{0, 1\}$          | Classical Perceptron                    |
| Sigmoid    | $(0, 1)$            | Binary Classification Output Layer      |
| Tanh       | $(-1, 1)$           | Hidden Layers (Traditional Networks)    |
| ReLU       | $[0, \infty)$       | Hidden Layers in CNNs and MLPs          |
| Leaky ReLU | $(-\infty, \infty)$ | Avoiding Dying ReLU                     |
| Softmax    | $(0, 1)$            | Multi-class Classification Output Layer |

**Note:** This experiment uses only the Step Activation Function. The remaining activation functions will be studied in subsequent experiments involving Multilayer Perceptrons and Deep Neural Networks.

## 5. Dataset Description

**Dataset:** Banknote Authentication Dataset

**Source:** UCI Machine Learning Repository

**Download Link:**

<https://archive.ics.uci.edu/dataset/267/banknote+authentication>

|                |                       |
|----------------|-----------------------|
| Instances      | 1372                  |
| Features       | 4 Numerical Features  |
| Classes        | 2                     |
| Missing Values | None                  |
| Task           | Binary Classification |

### Feature Description

- Variance
- Skewness
- Curtosis
- Entropy

Target Class

- 0 – Authentic Banknote
- 1 – Forged Banknote

## 6. Experimental Procedure

### Task 1: Dataset Exploration

- Load the dataset.
- Display the first five samples.
- Determine dataset dimensions.
- Identify missing values.
- Display descriptive statistics.

#### Expected Output

Dataset summary and statistical information.

### Task 2: Exploratory Data Analysis

Generate

- Histograms
- Correlation Heatmap
- Scatter Plot
- Boxplots

### Task 3: Data Preprocessing

- Normalize all numerical features.
- Split the dataset into Training (80%) and Testing (20%).

### Task 4: Perceptron Implementation

Implement from scratch

- Weight Initialization
- Bias Initialization
- Step Activation Function
- Forward Propagation
- Perceptron Learning Rule

### Task 5: Model Training

Display

- Epoch Number
- Misclassified Samples

- Updated Weights
- Updated Bias

### Task 6: Model Evaluation

Compute

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

## 7. Source Code

<https://github.com/ashrithaxx/Deep-Learning-Single-Layer-Perceptron>

## 8. Plots

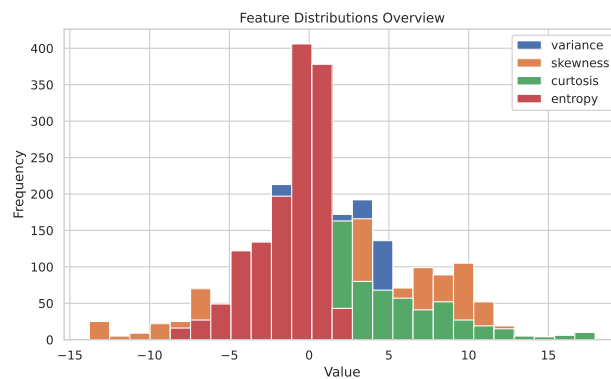


Figure 1: Feature Distribution

Interpretation: The diagram shows the distribution of the four features. The features are not perfectly symmetric, but there are no extreme outliers.

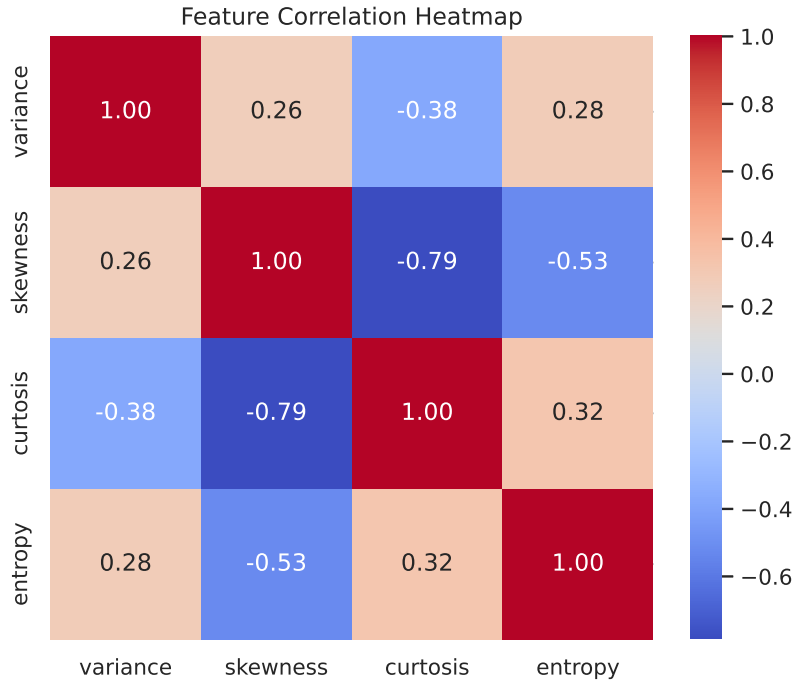


Figure 2: Feature Correlation Heatmap

Interpretation: The diagram shows the relationship between the features. Some are strongly positively or negatively correlated, while others are only weakly correlated.

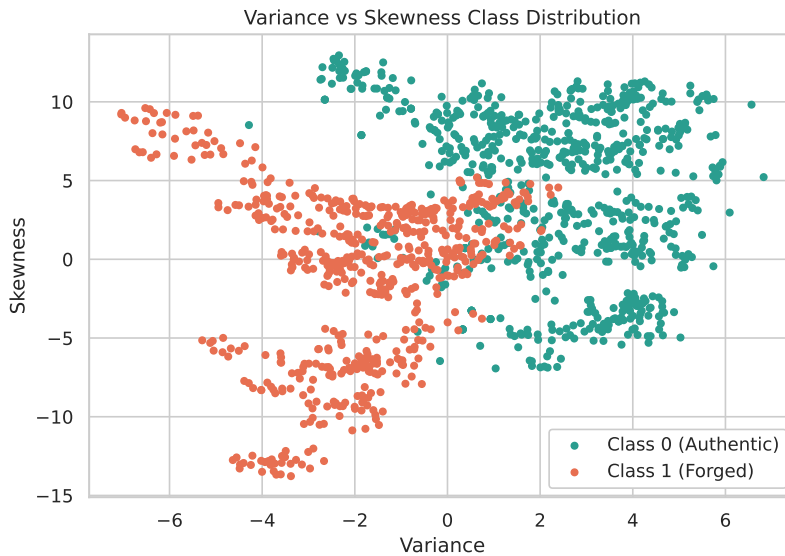


Figure 3: Variance vs Skewness Scatterplot

Interpretation: The diagram shows that the forged and authentic banknotes are mostly separated, with only a small overlap showing that the dataset is mostly linearly separable.

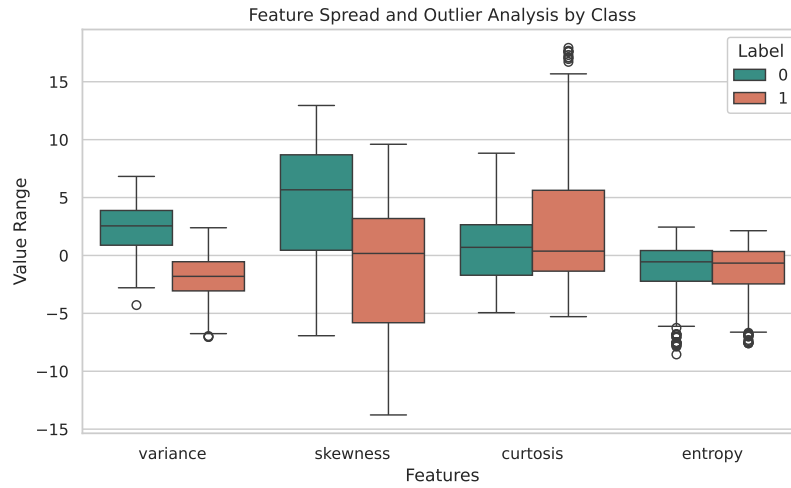


Figure 4: Feature Boxplots

Interpretation: The diagram shows the median, spread and highlights the possible outliers present. But the outliers do not dominate the dataset.



Figure 5: Training Errors vs Epoch

Interpretation: The training error decreases as the epochs increase and later stabilizes with only small fluctuations.

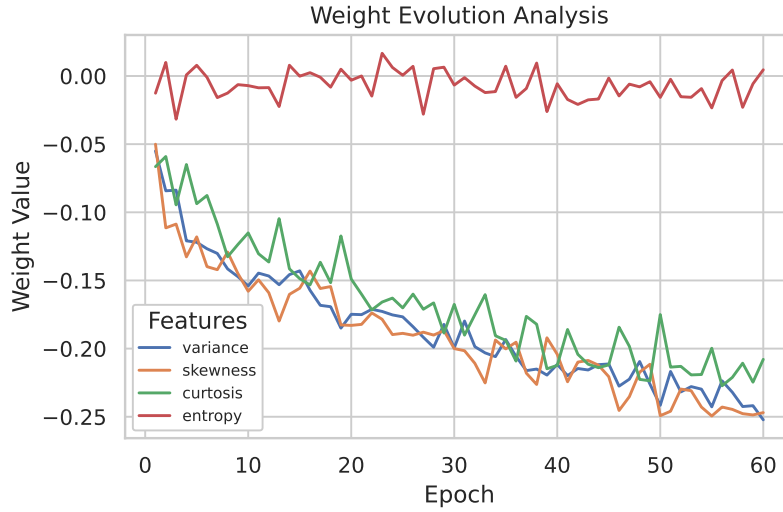


Figure 6: Weight Evolution

Interpretation: Initially the weights change significantly, but gradually stabilizes as the number of epochs increases. showing that the perceptron is learning and converging.

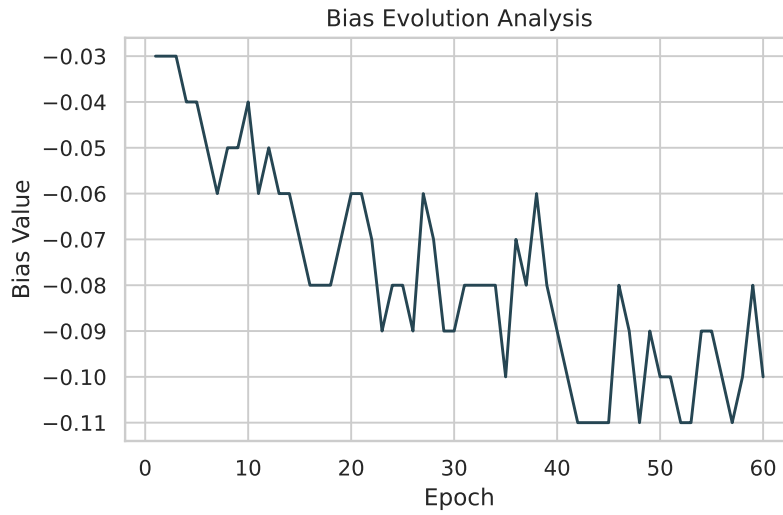


Figure 7: Bias Evolution

Interpretation: The bias is updated throughout training and remains stable despite small fluctuations showing that the perceptron has converged.



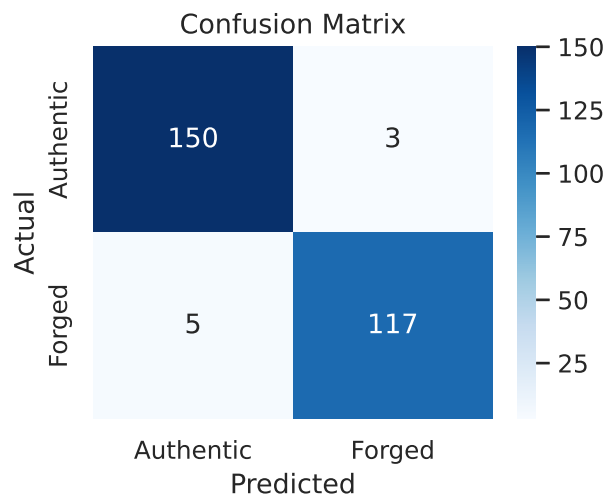


Figure 8: Confusion Matrix

Interpretation: The matrix shows that most banknotes are classified correctly (150 authentic and 117 forged), resulting in a high overall accuracy.

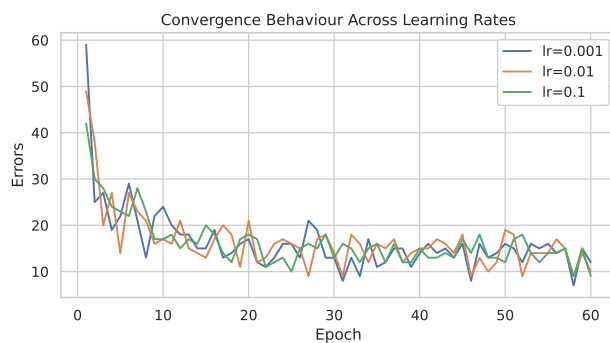


Figure 9: Learning Rate Comparison

Interpretation: Different learning rates affect convergence. larger learning rate fluctuates before stabilizing.

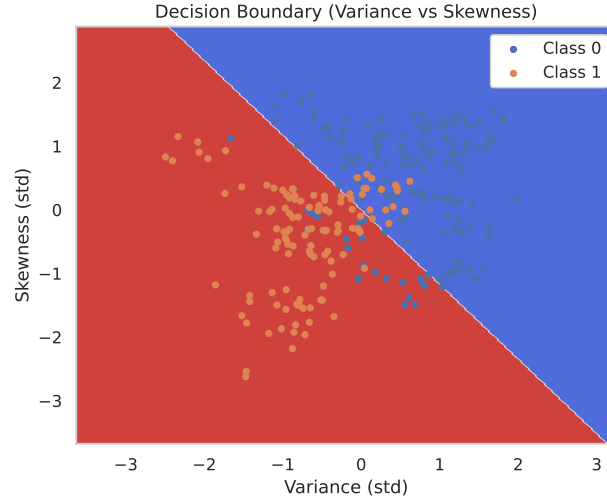


Figure 10: Decision Boundary

Interpretation: The diagram shows that authentic and forged banknotes are separated well based on variance and skewness with only a slight overlap.

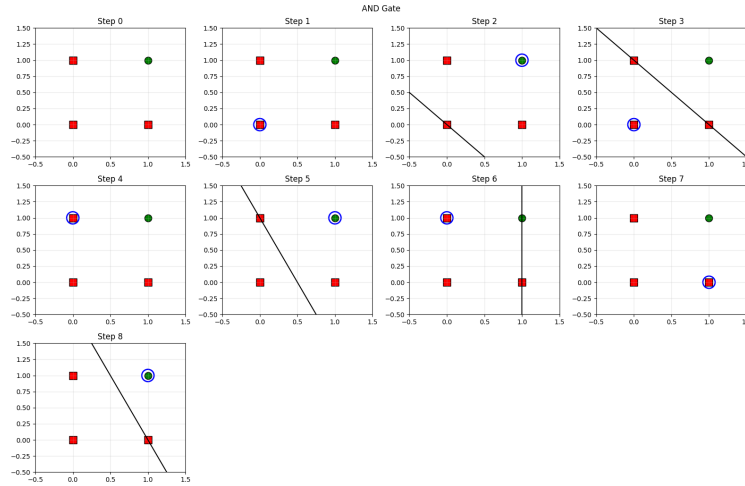


Figure 11: Decision Boundary for AND gate

Interpretation: The perceptron updates the decision boundary over time and converges by the final epoch thus correctly classifying the AND gate.

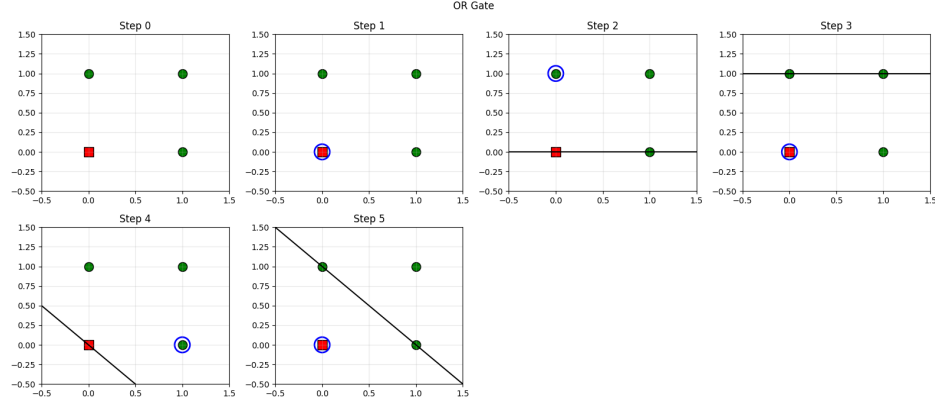


Figure 12: Decision Boundary for OR gate

Interpretation: The perceptron updates the decision boundary after each epoch. By the final iteration, it separates the negative output (0,0) from the three positive outputs, converging for the OR gate.

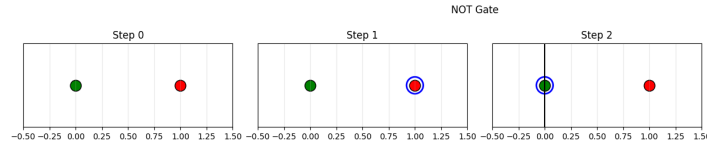


Figure 13: Decision Boundary for NOT gate

Interpretation: The decision boundary changes with each epoch. In the final step, the perceptron correctly classifies both inputs, showing successful learning of the NOT gate.

## 9. Performance Tables

### Training Summary

|                  |                                 |
|------------------|---------------------------------|
| Dataset Size     | 1372                            |
| Train/Test Split | 80%/20% (1097 Train / 275 Test) |
| Learning Rate    | 0.01                            |
| Epochs           | 60                              |
| Final Weights    | [-0.252 -0.247 -0.208 0.005]    |
| Final Bias       | -0.1000                         |
| Accuracy         | 0.97                            |
| Precision        | 0.97                            |
| Recall           | 0.96                            |
| F1-score         | 0.97                            |

### Epoch-wise Learning

| Epoch | Errors | Bias    | Weight 1 | Weight 2 | Weight 3 | Weight 4 |
|-------|--------|---------|----------|----------|----------|----------|
| 1     | 49     | -0.0300 | -0.055   | -0.050   | -0.067   | -0.013   |
| 10    | 17     | -0.0400 | -0.154   | -0.158   | -0.115   | -0.007   |
| 20    | 21     | -0.0600 | -0.175   | -0.183   | -0.149   | -0.003   |
| 30    | 14     | -0.0900 | -0.199   | -0.200   | -0.168   | -0.007   |
| 40    | 15     | -0.0900 | -0.212   | -0.204   | -0.212   | -0.006   |
| 50    | 19     | -0.1000 | -0.242   | -0.249   | -0.175   | -0.016   |
| 60    | 10     | -0.1000 | -0.252   | -0.247   | -0.208   | 0.005    |

## 10. Discussion

1. The step function gives 0 or 1 as the output whereas the sigmoid function gives any output from 0 to 1, ie, smooth output. As step function is not differentiable at the threshold, it cannot support learning in multilayer networks, and therefore is not used in modern deep learning models.
2. Both the manual as well as the Scikit-learn are based on the perceptron learning rule. However, Scikit-learn is more optimized and handles edge cases better.
3. When  $LR = 0.001$ , learning happens very slowly and larger no. of epochs is required to converge. When  $LR = 0.01$ , it reaches a stable result within a good no. of epochs. When  $LR = 0.1$ , convergence occurs faster, but there will be more fluctuations.
4. A single layer perceptron cannot handle XOR because it can only learn a linear decision boundary.
5. Without normalization, convergence is slower due to larger-range features. After normalization, the model converges faster and performs more accurately.

## 11. Report Contents

The report should include:

- Objective
- Background Theory
- Activation Functions
- Dataset Description
- Experimental Procedure
- Source Code
- Experimental Results
- Generated Plots with Interpretation

- Performance Tables
- Discussion
- Conclusion
- References

## 12. Conclusion

The Perceptron was trained and implemented manually on the Banknote Authentication dataset, with accuracy of 97% after 60 epochs showing that the data is linearly separable.

## 13. References

1. F. Rosenblatt, “The Perceptron,” *Psychological Review*, 1958.
2. I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*, MIT Press, 2016.
3. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
4. S. Haykin, *Neural Networks and Learning Machines*, Pearson, 2009.
5. UCI Machine Learning Repository – Banknote Authentication Dataset.
6. Scikit-learn Documentation: Perceptron.